Cover letter draft

Editor-in-Chief

Journal Editorial Office

Dear Editor,

Please consider our manuscript, “Calibration of Heart-Rate Reserve for Graded Cycle Ergometry: An Endpoint-Preserving Transfer Evaluated Against Linear Alternatives,” for publication as an Original Research Article.

This methods study addresses a practical limitation in heart-rate-based activity scoring. The INTERLIVE Network proposed transparent aggregation of time in 10% heart-rate-reserve clusters, but raw HRR is not a universal metabolic calibration. Using public graded-exercise data, we compared four candidate sport directions and several monotonic transformations while withholding the latest 30% of dated test files. We then locked a bounded, endpoint-preserving transfer and evaluated it in 84 temporally held-out cycling test files and an external PhysioNet cycling task with 18 participants.

Relative to uncalibrated HRR, the locked function reduced complete-test VO2-reserve mean absolute error by 1.07 percentage points in the temporal holdout and participant-level error by 1.38 points in the external task. Exact temporal 10% VO2-reserve-band agreement increased by 8.8 percentage points. The manuscript also reports the less favorable but important finding that the nonlinear function did not outperform development-fitted linear calibration for primary temporal VO2-reserve. Intensity-band and endpoint-exclusion analyses further distinguish the benefit of general group calibration from any unsupported claim of high-intensity curvature superiority.

The submission includes deterministic split labels, complete analysis code, an auditable source-data workbook, vector and 600-dpi figures, a 10-band implementation, strong linear comparisons, endpoint removal, intensity stratification, practical agreement, time-offset analysis, parameter sensitivity, and HR-anchor perturbation. The paper limits inference to graded cycling-intensity calibration and does not claim validation for fatigue, adaptation, injury, clinical outcomes, or wrist-device accuracy.

No new human data were collected. Both source datasets are public and deidentified, and their records report the original ethics approvals and consent processes. This work received no external funding, and the author declares no competing interests. The manuscript is not under consideration elsewhere and has not been published previously.

Thank you for considering this work. I would be pleased to respond to any questions.

Sincerely,

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